1(a)	$\mathrm{gf}(1) = \mathrm{g}\left(\frac{2(1)}{3\times1+1}\right) = 4-\left(\frac{2(1)}{3\times1+1}\right)^2$	M1
	$=\frac{15}{4}$ oe	A1
		(2)
(b)	$\mathrm{f}(x)\ldots0$ or $\mathrm{f}(x)<\frac{2}{3}$	B1
	0 ,, $\mathrm{f}(x)<\frac{2}{3}$	B1
		(2)

(c)	$y=\frac{2x}{3x+1}\Rightarrow 3xy+y=2x\Rightarrow 3xy-2x=-y\Rightarrow x(3y-2)=-y$ or $x=\frac{2y}{3y+1}\Rightarrow 3xy+x=2y\Rightarrow 3xy-2y=-x\Rightarrow y(3x-2)=-x$	M1
	$\Rightarrow \mathrm{f}^{-1}(x)=\frac{x}{2-3x}$	A1
		(2)
(d)	$\mathrm{f}^{-1}(x)=\mathrm{f}(x)$ **or** $\mathrm{f}^{-1}(x)=x$ **or** $\mathrm{f}(x)=x$ $"\frac{x}{2-3x}"=\frac{2x}{3x+1}$ **or** $"\frac{x}{2-3x}"=x$ **or** $\frac{2x}{3x+1}=x$ $\Rightarrow x(3x+1)=2x(2-3x)$ **or** $x=x(2-3x)$ **or** $2x=x(3x+1)$ $\Rightarrow x=...$	M1
	$x=0,\ \frac{1}{3}$	A1
		(2)
		Total 8

2(a)	$R = 25$	B1
	$\tan\alpha = \frac{24}{7} \Rightarrow \alpha = ...$	M1
	$\alpha = 1.287$	A1
		(3)
(b)(i)	$\text{Min} = \frac{5}{90 - 3\times"25"\times(-1)}$	M1
	$= \frac{1}{33}$	A1
(b)(ii)	$(2x + "1.287") = \pi, ... \Rightarrow x = ...$	M1
	$\Rightarrow x = \frac{\pi - "1.287"}{2} = 0.927$	A1
		(4)
		Total 7

3(a)	$\log_{10} y = \pm\frac{2}{3.5}\log_{10} x - 2$	M1
	$\log_{10} y = -\frac{4}{7}\log_{10} x - 2$	A1
		(2)

(b)	$\log_{10} y = -\frac{4}{7}\log_{10} x - 2 \Rightarrow$ $y = 10^{\left(-\frac{4}{7}\log_{10} x - 2\right)}$ or $y = 10^{\left(\log_{10} x^{-\frac{4}{7}} - 2\right)}$ or $y = 10^{-\frac{4}{7}\log_{10} x} \times 10^{-2}$ or e.g. $\log_{10} y = \log_{10} x^{-\frac{4}{7}} - 2 \Rightarrow \log_{10} \frac{y}{x^{-\frac{4}{7}}} = -2$ or e.g. $\log_{10} y = -\log_{10} x^{\frac{4}{7}} - 2 \Rightarrow \log_{10} yx^{\frac{4}{7}} = -2$	M1
	$y = 10^{\left(\log_{10} x^{-\frac{4}{7}} - 2\right)}$ or $y = 10^{-\frac{4}{7}\log_{10} x} \times 10^{-2} \Rightarrow y = x^{-\frac{4}{7}} \times 10^{-2}$ or e.g. $10^{\log_{10} \frac{y}{x^{-\frac{4}{7}}}} = 10^{-2} \Rightarrow \frac{y}{x^{-\frac{4}{7}}} = 10^{-2}$ or $10^{\log_{10} yx^{\frac{4}{7}}} = 10^{-2} \Rightarrow yx^{\frac{4}{7}} = 10^{-2}$	dM1
	$y = \frac{1}{100} x^{-\frac{4}{7}}$ oe	A1
		(3)
		Total 5

4(a)	$$\frac{49x}{x^2+x-12}+\frac{7x}{x+4}=\frac{49x+7x(x-3)}{(x+4)(x-3)}$$ or e.g. $$\frac{49x(x+4)}{(x^2+x-12)(x+4)}+\frac{7x(x^2+x-12)}{(x+4)(x^2+x-12)}=\frac{49x(x+4)+7x(x^2+x-12)}{(x+4)(x^2+x-12)}$$	M1A1
	$$\frac{49x+7x(x-3)}{(x+4)(x-3)}=\frac{49x+7x^2-21x}{(x+4)(x-3)}=\frac{7x^2+28x}{(x+4)(x-3)}$$ $$=\frac{7x\cancel{(x+4)}}{\cancel{(x+4)}(x-3)}=\frac{7x}{(x-3)}*$$	A1*
		(3)

(b)	$f(x)=\frac{7x}{(x-3)} \Rightarrow f'(x)=\frac{(x-3)\times 7-7x\times 1}{(x-3)^2}$	M1
	$f(x)=\frac{7x}{(x-3)} \Rightarrow f'(x)=-\frac{21}{(x-3)^2}$	A1
		(2)
(b) ALT	$f(x)=\frac{7x}{(x-3)}=7+\frac{21}{x-3} \Rightarrow f'(x)=-\frac{21}{(x-3)^2}$	M1A1
		Total 5

5(i)	$\cos 6x = 1 - 2\sin^2 3x \Rightarrow \sin^2 3x = \frac{1}{2} - \frac{1}{2}\cos 6x$ $\int \sin^2 3x \, dx = \int \left(\frac{1}{2} - \frac{1}{2}\cos 6x \right) dx$	M1
	$= \frac{1}{2}x - \frac{1}{12}\sin 6x (+c)$	A1
		(2)
(ii)	$\int x\left(x^2+4\right)^{\frac{3}{2}} dx = \frac{1}{5}\left(x^2+4\right)^{\frac{5}{2}} (+c)$	M1A1
		(2)
		Total 4

6(a)	$\theta = 75,\ t = 0 \Rightarrow 75 = 21 + A \Rightarrow A = ...$	M1
	$A = 54$	A1
		(2)
(b)	$\theta = 21 + 54\mathrm{e}^{-kt} \Rightarrow 25 = 21 + 54\mathrm{e}^{-5k}$	M1
	$54\mathrm{e}^{-5k} = 4 \Rightarrow \mathrm{e}^{-5k} = \frac{2}{27} \Rightarrow -5k = \ln\frac{2}{27} \Rightarrow k = ...$	M1
	$k = -\frac{1}{5}\ln\frac{2}{27} = 0.521$	A1
		(3)

(c)	$\theta = 21 + 54\mathrm{e}^{-0.521t} \Rightarrow \dfrac{\mathrm{d}\theta}{\mathrm{d}t} = -28.1...\mathrm{e}^{-0.521t}$	M1
	$-28.1...\mathrm{e}^{-0.521T} = -9 \Rightarrow \mathrm{e}^{-0.521T} = \dfrac{9}{28.1...} \Rightarrow -0.521T = \ln\left(\dfrac{9}{28.1...}\right)$ $\Rightarrow T = \ln\left(\dfrac{9}{28.1...}\right) \div -0.521$	dM1
	$= 2.19$	A1
		(3)

7(a)	$y = \mathrm{e}^{-x^2}\sin 3x \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = 3\mathrm{e}^{-x^2}\cos 3x - 2x\mathrm{e}^{-x^2}\sin 3x$	M1A1
	$3\mathrm{e}^{-x^2}\cos 3x - 2x\mathrm{e}^{-x^2}\sin 3x = 0 \Rightarrow 3\cos 3x - 2x\sin 3x = 0$ $\Rightarrow 3\cos 3x = 2x\sin 3x \Rightarrow \tan 3x = \dfrac{3}{2x}$	dM1
	$x = \dfrac{1}{3}\arctan\left(\dfrac{3}{2x}\right)$ *	A1*
		(4)
(b)(i)	$x_1 = 0.4 \Rightarrow x_2 = \dfrac{1}{3}\arctan\left(\dfrac{3}{2\times 0.4}\right)$	M1
	$\left(x_2 =\right) 0.4367$	A1
(ii)	$\left(x_4 =\right) 0.4307$	A1
		(3)

(c)	e.g. $\mathrm{f}(x) = x - \frac{1}{3}\arctan\left(\frac{3}{2x}\right)$ $\mathrm{f}(0.4305) = 0.4305 - \frac{1}{3}\arctan\left(\frac{3}{2\times 0.4305}\right)\left(= 6.38\times 10^{-5}\right)$ $\mathrm{f}(0.4295) = 0.4295 - \frac{1}{3}\arctan\left(\frac{3}{2\times 0.4295}\right)\left(= -1.141...\times 10^{-3}\right)$ or e.g. $\mathrm{f}(x) = 3\mathrm{e}^{-x^2}\cos 3x - 2x\mathrm{e}^{-x^2}\sin 3x$ $\mathrm{f}(0.4305) = 3\mathrm{e}^{-(0.4305)^2}\cos 3(0.4305) - 2(0.4305)\mathrm{e}^{-(0.4305)^2}\sin 3\ (0.4305)\left(= -4.968...\times 10^{-4}\right)$ $\mathrm{f}(0.4295) = 3\mathrm{e}^{-(0.4295)^2}\cos 3(0.4295) - 2(0.4295)\mathrm{e}^{-(0.4295)^2}\sin 3\ (0.4295)\left(= 8.885...\times 10^{-3}\right)$	M1
	Sign change therefore x is 0.430 to 3dp	A1
		(2)
		Total 9

8(a)	$\tan 3x \equiv \tan(2x+x) \equiv \dfrac{\tan 2x + \tan x}{1-\tan 2x \tan x}$ or e.g. $\tan 3x \equiv \dfrac{\sin 3x}{\cos 3x} \equiv \dfrac{\sin(2x+x)}{\cos(2x+x)} \equiv \dfrac{\sin 2x\cos x + \cos 2x \sin x}{\cos 2x \cos x - \sin 2x \sin x} \equiv \dfrac{\tan 2x + \tan x}{1-\tan 2x \tan x}$	M1
	$\dfrac{\tan 2x + \tan x}{1-\tan 2x \tan x} \equiv \dfrac{\dfrac{2\tan x}{1-\tan^2 x} + \tan x}{1-\dfrac{2\tan x}{1-\tan^2 x}\tan x}$	dM1
	$\dfrac{\dfrac{2\tan x}{1-\tan^2 x} + \tan x}{1-\dfrac{2\tan x}{1-\tan^2 x}\tan x} \equiv \dfrac{2\tan x + \tan x - \tan^3 x}{1-\tan^2 x - 2\tan^2 x}$ $\dfrac{3\tan x - \tan^3 x}{1-3\tan^2 x}$ *	A1*
		(3)

Way 2	$\tan 3x \equiv \dfrac{\sin 3x}{\cos 3x} \equiv \dfrac{\sin(2x+x)}{\cos(2x+x)} \equiv \dfrac{\sin 2x\cos x+\cos 2x\sin x}{\cos 2x\cos x-\sin 2x\sin x}$ $\equiv \dfrac{2\sin x\cos^2 x+(\cos^2 x-\sin^2 x)\sin x}{(\cos^2 x-\sin^2 x)\cos x-2\sin^2 x\cos x}$	M1
	$\equiv \dfrac{2\sin x\cos^2 x+\cos^2 x\sin x-\sin^3 x}{\cos^3 x-\sin^2 x\cos x-2\sin^2 x\cos x} \equiv \dfrac{3\sin x\cos^2 x-\sin^3 x}{\cos^3 x-3\sin^2 x\cos x}$	dM1
	$\equiv \dfrac{3\tan x-\tan^3 x}{1-3\tan^2 x}$ *	A1*

(b)	$\dfrac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}=2\sec^2 3\theta-8\Rightarrow\tan 3\theta=2\sec^2 3\theta-8$ $\Rightarrow\tan 3\theta=2\left(1+\tan^2 3\theta\right)-8$	M1
	$\Rightarrow 2\tan^2 3\theta-\tan 3\theta-6=0$ or e.g. $\Rightarrow 2\tan^2 3\theta-\tan 3\theta=6$	A1
	$\left(2\tan 3\theta+3\right)\left(\tan 3\theta-2\right)=0\Rightarrow\tan 3\theta=-\dfrac{3}{2},\ 2$ $\tan 3\theta=-\dfrac{3}{2}\Rightarrow 3\theta=...\Rightarrow\theta=...$ or $\tan 3\theta=2\Rightarrow 3\theta=...\Rightarrow\theta=...$	dM1
	$\theta=0.37,\ 0.72,\ 1.42$	A1A1
		(5)
		Total 8

9(a)	$x = 0 \Rightarrow \frac{2}{3}\sin\left(3y + \frac{\pi}{4}\right) = 0 \Rightarrow 3y + \frac{\pi}{4} = \pi,\ 2\pi, \ldots \Rightarrow y = \ldots$	M1
	$y = \frac{\pi}{4}$ or $y = \frac{7\pi}{12}$	A1
	$y = \frac{\pi}{4}$ and $y = \frac{7\pi}{12}$	A1
		(3)

(b)	$x=\frac{2}{3}\sin\left(3y+\frac{\pi}{4}\right)\Rightarrow\frac{\mathrm{d}x}{\mathrm{d}y}=\frac{2}{3}\times3\cos\left(3y+\frac{\pi}{4}\right)\left(=2\cos\left(3y+\frac{\pi}{4}\right)\right)$	B1
	$\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2\cos\left(3y+\frac{\pi}{4}\right)}\Rightarrow\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2=\frac{1}{4\cos^2\left(3y+\frac{\pi}{4}\right)}=\frac{1}{4\left(1-\sin^2\left(3y+\frac{\pi}{4}\right)\right)}$	M1
	$=\frac{1}{4\left(1-\left(\frac{3x}{2}\right)^2\right)}$	**d**M1
	$=\frac{1}{4-9x^2}$	A1
		(4)

(b) Way 2	$x = \frac{2}{3}\sin\left(3y + \frac{\pi}{4}\right) \Rightarrow 3y + \frac{\pi}{4} = \sin^{-1}\frac{3x}{2}$	B1
	$3y + \frac{\pi}{4} = \sin^{-1}\frac{3x}{2} \Rightarrow 3\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-\frac{9x^2}{4}}} \times \frac{3}{2}$	M1
	$3\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-\frac{9x^2}{4}}} \times \frac{3}{2} \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = \left(\frac{1}{2\sqrt{1-\frac{9x^2}{4}}}\right)^2$	**d**M1
	$= \frac{1}{4-9x^2}$	A1

(c)	Identifies that one of the gradients is $\sqrt{"4"}$ or that one of the gradients is $\frac{1}{\sqrt{"4"}}$ **or** Identifies that one of the gradients is 2 or that one of the gradients is $\frac{1}{2}$	B1ft
	At A: $y-"\frac{\pi}{4}"="2"x$ **and** At B: $y-"\frac{7\pi}{12}"="\frac{1}{2}"x$	M1
	$"2"x+"\frac{\pi}{4}"="\frac{1}{2}"x+"\frac{7\pi}{12}"\Rightarrow x=...$	**d**M1
	$x=\frac{2\pi}{9}$	A1
		(4)
		Total 11

10(a)(i)	$(y=)10+k$	B1
(ii)	$x=\frac{10}{k}$ **or** $y=k$	B1
	$x=\frac{10}{k}$ **and** $y=k$	B1
		(3)

(b)	$kx-10+k=2k \Rightarrow x=...$ **or** $-kx+10+k=2k \Rightarrow x=...$	M1
	$kx-10+k=2k \Rightarrow x=...$ **and** $-kx+10+k=2k \Rightarrow x=...$	dM1
	$x \,,, \dfrac{10-k}{k}$ or $x \ldots \dfrac{10+k}{k}$ oe	A1
		(3)

	(b) Alternative by squaring.	
	$\lvert kx-10\rvert+k=2k \Rightarrow \lvert kx-10\rvert=k \Rightarrow k^2x^2-20kx+100=k^2$ $\Rightarrow k^2x^2-20kx+100-k^2=0$	M1
	$k^2x^2-20kx+100-k^2=0 \Rightarrow x=\dfrac{20k\pm\sqrt{400k^2-4k^2\left(100-k^2\right)}}{2k^2}$ $=\dfrac{10\pm k}{k}$	dM1
	$x \,,, \dfrac{10-k}{k}$ or $x \ldots \dfrac{10+k}{k}$ oe	A1

(c)	$k > 3$	B1
	Maximum value of k occurs when $y = 3x + 1$ passes through vertex e.g. when $"k" = 3\times"\frac{10}{k}"+1 \Rightarrow k = ...$ **See below for alternative approaches.**	M1
	$k^2 - k - 30 = 0 \Rightarrow (k+5)(k-6) = 0 \Rightarrow k = 6$ So **maximum** k is 6	A1
	$3 < k < 6$	A1
		(4)
		Total 10